# Introduction to Lie Group Theory

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# Contents

# Chapter 1

# Matrix Groups

## 1.1 Groups of Matrices

Let $M_{m,n}(\mathbb{K})$ be the set of $m \times n$ matrices whose entries are in $\mathbb{K}$, a commutative field. We will also use the special notations $M_n(\mathbb{K}) = M_{n.n}(\mathbb{K})$, and $\mathbb{K}^n = M_{n,1}(\mathbb{K})$.

$M_{m,n}(\mathbb{K})$ is a vector space over $\mathbb{K}$ with dimension $mn$. As well as being a vector space over $\mathbb{K}$ with dimension $n^2$, $M_n(\mathbb{K})$ is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensinal algebra over $\mathbb{K}$.

We will use the notation

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible $n \times n$ matrices, known as the $n \times n$ *general linear group*, and

$$SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq GL_n(\mathbb{K})$$

for the set of $n \times n$ unimodular matrices, known as the $n \times n$ *special linear group*.

**Theorem 1.1.** *The sets* $GL_n(\mathbb{K})$ *and* $SL_n(\mathbb{K})$ *are groups under matrix multiplication. Furthermore,* $SL_n(\mathbb{K})$ *is a subgroup of* $GL_n(\mathbb{K})$*.*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continous.

**Proposition 1.1.** *Let* $\mathbb{K} = \mathbb{R}$ *or* $\mathbb{C}$. *Then*

(i) $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ *is an open subset;*

(ii) $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ *is a closed subset.*

*Proof.* $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$, which is open since $\{0\}$ is closed. Similarly, $SL_n(\mathbb{K}) = \det^{-1}\{1\}$, which is closed since the singleton $\{1\}$ is closed in $\mathbb{K}$. ∎

**Definition 1.1.** Let G be a topological space, and view $G \times G$ as the product space. Suppose that G is also a group with multiplication map mult : $G \times G \longrightarrow G$, and inverse map inv : $G \longrightarrow G$. G is a *topological group* if mult and inv are continous.

**Theorem 1.2.** *For* $\mathbb{K} = \mathbb{R}$ *or* $\mathbb{C}$, $GL_n(\mathbb{K})$ *and* $SL_n(\mathbb{K})$ *are topological groups.*

## 1.2 Matrix Groups

**Definition 1.2.** A subgroup G of $GL_n(\mathbb{K})$ is known as a *matrix group*, given it is a closed subset of $GL_n(\mathbb{K})$.

*Remark.* We may consider $GL_n(\mathbb{K})$ as a subgroup of $GL_{n+1}(\mathbb{K})$ by identifying the $n \times n$ matrix $A = [a_{ij}]$ with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that $GL_n(\mathbb{K})$ is closed in $GL_{n+1}(\mathbb{K})$. With the aid of this embedding, any matrix subgroup of $GL_n(\mathbb{K})$ can also be viewed as a matrix subgroup of $GL_{n+1}(\mathbb{K})$.

**Definition 1.3.** Let G, H be two matrix groups. A groups homomorphism $\varphi : G \longrightarrow H$ is a *continuous homomorphism of matrix groups* if it is continuous and its image $\varphi(G)$ is a closed subgroup of H.

**Examples.** Some important examples of real and complex matrix groups:

1. **Groups of Upper Triangular Matrices**

   For $n \geqslant 1$, an $n \times n$ matrix $A = [a_{ij}]$ is said to be *upper triangular* if $a_{ij} = 0$ if $i < j$. An upper traingular matrix is said to be *unipotent* of all of it's diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of $\mathrm{GL}_n(\mathbb{K})$ is

$$\mathrm{UT}_n(\mathbb{K}) = \{A \in \mathrm{GL}_n(\mathbb{K}) : \text{A is upper triangular}\},$$

while the *unipotent* subgroup of $\mathrm{GL}_n(\mathbb{K})$ is

$$\mathrm{SUT}_n(\mathbb{K}) = \{A \in \mathrm{GL}_n(\mathbb{K}) : \text{A is unipotent}\}.$$

It is easy to verify that $\mathrm{UT}_n(\mathbb{K})$ and $\mathrm{SUT}_n(\mathbb{K})$ are closed subgroups of $\mathrm{GL}_n(\mathbb{K})$, and that $\mathrm{SUT}_n(\mathbb{K})$ is a subgroup of $\mathrm{UT}_n(\mathbb{K})$.

2. **Affine Groups**

   The *n-dimensional affine group* over $\mathbb{K}$ is

$$\mathrm{Aff}_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in \mathrm{GL}_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

   which is clearly a closed subgroup of $\mathrm{GL}_{n+1}(\mathbb{K})$. If we identify $x \in \mathbb{K}^n$ with $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$, then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

   we obtain an action of $\mathrm{Aff}_n(\mathbb{K})$ on $\mathbb{K}^n$.

   The group $\mathrm{Aff}_n(\mathbb{K})$ can be expressed as the semi-direct product of $\mathrm{Trans}_n(\mathbb{K})$ and $\mathrm{GL}_n(\mathbb{K})$, where

$$\mathrm{Trans}_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

   is a closed normal subgroup of $\mathrm{Aff}_n(\mathbb{K})$. That is,

$$\mathrm{Aff}_n(\mathbb{K}) = \mathrm{GL}_n(\mathbb{K}) \ltimes \mathrm{Trans}_n(\mathbb{K}).$$

3. **Orthagonal Groups**

   The $n \times n$ *real orthagonal group* is the subset

$$\mathrm{O}(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) : A^T A = I_n\},$$

   where $A^T$ is the transpose of $A = [a_{ij}]$ whose entris are given by $(A^T)_{ij} = a_{ji}$. $\mathrm{O}(n)$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{R})$. For all $A \in \mathrm{O}(n)$, since $\det A = \det A^T$, $\det A = \pm 1$. The subgroup

$$\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}$$

is known as the *special orthagonal group*. One of the main reasons for the study of orthagonal groups is due to their relation with isometries. If $A \in GL_n(\mathbb{R})$, then it can be proved that $A$ is an linear isometry *iff* $A$ is orthagonal. The elements of $SO(n)$ are called *direct isometries* or *rotations*.

4. **Unitary Groups**

   The $n \times n$ *unitary group* is the subset

   $$U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I_n\},$$

   where $A^*$ is the *hermitian conjugate* of $A$, that is $A^*)_{ij} = a_{ji}$. $U(n)$ is clearly a closed subgroup of $GL_n(\mathbb{C})$. The *special unitary group* is the subset

   $$SU(n) = \{A \in U(n) : \det A = 1\},$$

   which is also a normal subgroup of $U(n)$.

5. **Symplectic Groups**

   A matrix $S$ is called *skew symmetric* if $S^T = -S$. The standard example of this is built up using the $2 \times 2$ block

   $$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

   When $m \geqslant 1$ we have the non degenrative skew symmetric matrix

   $$J_{2m} = \begin{bmatrix} J & O_2 & \cdots & O_2 \\ O_2 & J & \cdots & O_2 \\ \vdots & \vdots & \ddots & \vdots \\ O_2 & O_2 & \cdots & J \end{bmatrix}.$$

   The matrix group

   $$\mathrm{Symp}_{2m}(\mathbb{R}) = \{A \in GL_{2m}(\mathbb{R}) : A^T J_{2m} A = J_{2m}\}$$

   is called the $2m \times 2m$ *real symplectic group*. It can be proved that if $A \in \mathrm{Symp}_{2m}(\mathbb{R})$, then $\det A = 1$.

## 1.3 Complex Matrices as Real Matrices

Similar to how complex numbers can be viewed as a 2-dimensional real vector space with basis $(1, i)$, every $n \times n$ complex matrix can also be viewed as a $2n \times 2n$ real matrix using the following construction or the ones similar to it.

We identify each complex number $z = x + iy$ with a $2 \times 2$ matrix by defining a funciton

$$\rho : \mathbb{C} \longrightarrow M_2(\mathbb{R}); \quad \rho(x + iy) = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

This can also be expressed as

$$\rho(x + iy) = xI_2 - yJ.$$

As $\rho$ is an injective ring homomorphism, we can view $\mathbb{C}$ as a subring of $M_2(\mathbb{R})$. Notice that $\rho(\bar{z}) = \rho(z)^T$.

For a $n \times n$ complex matrix $Z$, we define a function $\rho_n : M_n(\mathbb{C}) \longrightarrow M_{2n}(\mathbb{R})$;

$$\rho_n(Z) = \begin{bmatrix} \rho(z_{11}) & \rho(z_{12}) & \cdots & \rho(z_{1n}) \\ \rho(z_{21}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \rho(z_{n1}) & \cdots & \cdots & \rho(z_{nn}) \end{bmatrix}$$

Then $\rho_n$ is an continuous injective ring homomorphism. Generally for $n \times n$ real matrices $X$ and $Y$,

$$\rho_n(X + iY) = \rho_n(X) - J_{2n}\rho_n(Y)J_{2n}.$$

## 1.4 Matrix Groups of Normed Vector Spaces

It is often desireable to generalize the notion of matrix spaces to srbitrary finite dimensional real or complex vector spaces. Here we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

**Theorem 1.3.** *Let $(U, \mu)$ and $(V, \nu)$ be two finite dimensional normed vector spaces of the same dimension over $\mathbb{K}$. Then any linear isomorphism $\varphi : U \longrightarrow V$ is a homeomorphism.*

**Corollary.** *If $\mu$ and $\nu$ are to norms on a finite dimensional vector space, then they give raise to the same topology.*

*Proof.* The identity function is a linear isomorphism, which must be continuous. This implies that every open set in the topology of $\nu$ is open in the topology of $\mu$ and vice-versa. ■

Given a finite dimensional normed vector space $V, \nu$ and a linear transformation $\alpha : V \longrightarrow V$, we define the *operator norm* of $\alpha$ with respect to $\nu$ by

$$||\alpha||_\nu = \mathrm{sup}\{\nu(\alpha(x)) : x \in V, \nu(x) = 1\}$$

This defines a norm on $\mathrm{End}_{\mathbb{K}}(V)$, the set of all linear transformations $V \longrightarrow V$, which makes it a finite dimensional normed vector space overe $\mathbb{K}$.

**Proposition 1.2.** *If $\mu$ and $\nu$ are two norms on a finite dimensional vector space $V$, then the topologies associated to $\mathrm{End}_{\mathbb{K}}(V)$ with respect to $\mu$ and $\nu$ are the same.*

Inside $\mathrm{End}_{\mathbb{K}}(V)$ we have the group of linear isomorphisms, $\mathrm{GL}_{\mathbb{K}}(V) \subseteq \mathrm{End}_{\mathbb{K}}(V)$, the *general linear group* of the vector space $V$.

**Proposition 1.3.** *Let $U$ and $V$ be two vector spaces of the same dimension. If $\varphi : U \longrightarrow V$ is a linear isomorphism, then it induces a continuous group isomorphism with a continuous inverse,*

$$\varphi_* : \mathrm{GL}_{\mathbb{K}}(U) \longrightarrow \mathrm{GL}_{\mathbb{K}}(V); \quad \varphi_*(\alpha) = \varphi \circ \alpha \circ \varphi^{-1}$$

Generally, given a finite basis for $V$, there is an associated linear isomorphism $V \cong \mathbb{K}^n$ for some $n$, a ring isomorphism $\mathrm{End}_{\mathbb{K}}(V) \cong M_n(\mathbb{K})$ and a group isomorphism $\mathrm{GL}_{\mathbb{K}}(V) \cong \mathrm{GL}_n(\mathbb{K})$.

**Theorem 1.4.** *Let $V$ be a finite dimensional vector space over $\mathbb{K}$ with dimension $n$ and let $\varphi : V \longrightarrow \mathbb{K}^n$ be a linear isomorphism. Then $\varphi_* : \mathrm{GL}_{\mathbb{K}}(V) \longrightarrow \mathrm{GL}_n(\mathbb{K})$ is a continuous grouop isomorphism with a continuous inverse. Hence $\mathrm{GL}_{\mathbb{K}}(V)$ is a matrix group.*

## 1.5 Matrix Exponential and Logarithm

We define the exponential and logarithm functions for matrices using their power series. Throught this section, we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

The *exponential function* is, $\exp : M_n(\mathbb{K}) \longrightarrow \mathrm{GL}_n(\mathbb{K})$;

$$\exp(A) = \sum_{n \geqslant 0} \frac{1}{n!} A^n = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$$

**Proposition 1.4.** *Let* $A \in M_n(\mathbb{K})$*.*

(i) *For* $u, v \in \mathbb{C}, \exp((u+v)A) = \exp(uA)\exp(vA)$*.*

(ii) $\exp(A)^{-1} = \exp(-A)$*.*

(iii) *If* $B \in M_n(\mathbb{K})$ *commutes with* $A$*, then* $\exp(A+B) = \exp(A)\exp(B)$*.*

We define the it logarithm function by

$$\log : N_{M_n(\mathbb{K})}(I;1) \longrightarrow M_n(\mathbb{K});$$

$$\log(A) = \sum_{n \geqslant 1} \frac{(-1)^{n-1}}{n}(A-I)^n$$

**Proposition 1.5.** *The* exp *and* log *functions satisfy*

(i) *if* $\|A - I\| < 1$*, then* $\exp(\log(A)) = A$*;*

(ii) *if* $\|\exp(B) - I\| < 1$*, then* $\log(\exp(B)) = B$*.*

The functions exp and log are continuous and in fact infinitely differentiable on their domains. By continuity of exp at $O$, there is a $\delta_1 > 0$ such that

$$N_{M_n(\mathbb{K})}(O;r) \subseteq \exp^{-1} N_{GL_n(\mathbb{K})}(I;1).$$

Since $\exp N_{M_n(\mathbb{K})}(O;r) \subseteq N_{M_n(\mathbb{K})}(I; e^r - 1)$, we can take $\delta_1 = \log 2$. Hence we have the following result.

**Proposition 1.6.** *The exponential function* exp *is injective when restricted to the open subset* $N_{M_n(\mathbb{K})}(O; \log 2) \subseteq M_n(\mathbb{K})$*. Hence it is locally a diffeomorphism at* $O$ *with the local inverse* log*.*

## 1.6 Differential Equations in Matrices

**Definition 1.4.** A *differentiable curve* in $M_n(\mathbb{K})$ is a function $\alpha : (a,b) \longrightarrow M_n(\mathbb{K})$ for which the derivative $\alpha'(t)$ exists for each $t \in (a.b)$. Here $\alpha'(t)$ is defined as an element of $M_n(\mathbb{K})$ by

$$\alpha'(t) = \lim_{s \to t} \frac{1}{(s-t)}(\alpha(s) - \alpha(t)),$$

provided that the limit exists.

Consider the first order differential equation $\alpha'(t) = \alpha(t)A$, where $\alpha$ is assumed to be a differentiable curve.

**Theorem 1.5.** *For* $A, C \in M_n(\mathbb{R})$ *with* $A$ *non-zero, and* $a < 0 < b$*, the differential equation* $\alpha'(t) = \alpha(t)A$*, has a unique solution* $\alpha : (a.b) \longrightarrow M_n(\mathbb{R})$ *for which* $\alpha(0) = C$*. Furthermore, if* $C$ *is invertible, then so is* $\alpha(t)$ *for* $t \in (a, b)$*.*

*Proof.* Consider the function $\alpha : (a, b) \longrightarrow M_n(\mathbb{R})$; $\alpha(t) = \exp(tA)$. It is differentiable with

$$\alpha'(t) = \sum k \geqslant 1 \frac{t^{k-1}}{(k-1)!} A^k = A \exp(tA)$$

Hence $\alpha$ satisfies the differential equation with boundary condition $\alpha(0) = I$. Clearly, $\alpha(t) = C \exp(tA)$ is one solution to $\alpha(0) = 0$. If $\beta$ is another such solution, then $\gamma(t) = \beta(t) \exp(-tA)$ satisfies

$$\begin{aligned} \gamma'(t) &= \beta'(t) \exp(-tA) + \beta(t) \exp(-tA)(-A) \\ &= \beta(t)A \exp(-tA) - \beta(t)A \exp(-tA) \\ &= O. \end{aligned}$$

Hence $\gamma$ is a constant function with $\gamma(t) = \gamma(0) = C$. Thus $\beta(t) = C \exp(tA) = \alpha(t)$. ■

## 1.7 One Parameter Subgroups in Matrix Groups

**Definition 1.5.** A *differentiable curve* is $G$ is a function $\gamma : (a, b) \longrightarrow G$ for which derivative $\gamma'(t)$ exists at each $t \in (a.b)$.

We usually assume that $a < 0 < b$ when considering such curves. Now suppose that $\varepsilon > 0$ or $\varepsilon = \infty$.

**Definition 1.6.** A *one-parameter subgroup* in $G$ is a continuous function $\gamma : (-\varepsilon, \varepsilon) \longrightarrow G$ which is differentiable at 0 and also satisfies $\gamma(s + t) = \gamma(s)\gamma(t)$ whenever $s, t, s + t \in (-\varepsilon, \varepsilon)$. If $\varepsilon = \infty$, then $\gamma : \mathbb{R} \longrightarrow G$ is called a *one-parameter subgroup* of $G$.

**Proposition 1.7.** *Let* $\gamma : (-\varepsilon, \varepsilon) \longrightarrow G$ *be a one-parameter semigroup. Then for every* $t \in (-\varepsilon, \varepsilon)$*,* $\gamma$ *is differentiable at* $t$*, and*

$$\gamma'(t) = \gamma'(0)\gamma(t) = \gamma(t)\gamma(0).$$

**Proposition 1.8.** *Let* $\gamma : (-\varepsilon, \varepsilon) \longrightarrow G$ *be a one-parameter semigroup. Then there is a unique extension to a one-parameter group* $\tilde{\gamma} : \mathbb{R} \longrightarrow G$.

*Proof.* Let $t \in \mathbb{R}$. Then for a large enough natural number $m$, $t/m \in (-\varepsilon, \varepsilon)$, hence $\gamma(t/m), \gamma(t/m)^m \in G$. Similarly for a second natural number $n$, $\gamma(t/n), \gamma(t/n)^n \in G$. Since $mn \geqslant m$ and $mn \geqslant n$, we also have $t/mn \in (-\varepsilon, \varepsilon)$. Therefore,

$$\begin{aligned} \gamma(t/n)^n &= \gamma(mt/mn) \\ &= \gamma(t/mn)^{mn} \\ &= \gamma(nt/mn)^m \\ &= \gamma(t/m)^m \end{aligned}$$

Thus $\gamma(t/n)^n = \gamma(t/m)^m$. This defines a function

$$\tilde{\gamma} : \mathbb{R} \longrightarrow G; \quad \tilde{\gamma}(t) = \gamma(t/n)^n$$

for large n. It is easy to see that $\tilde{\gamma}$ is an one-parameter subgroup in G. ■

**Theorem 1.6.** *Let* $\gamma : \mathbb{R} \longrightarrow G$ *be a one-parameter group in* $G$*. Then it has the form* $\gamma(t) = \exp(tA)$ *for some* $A \in M_n(\mathbb{K})$.

*Proof.* Let $A = \gamma'(0)$. Since $\gamma$ satisfies the differential equation $\gamma'(t) = \gamma(t)A$, $\gamma(0) = A$, by Theorem 1.5, this equation has a unique solution $\gamma(t) = \exp(tA)$. ■

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# Chapter 2

# Algebras and Quaternions

## 2.1 Algebras

**Definition 2.1.** A *finite dimensional algebra* $A$ is a finite dimesional vecotr space over $\mathbb{K}$ which is also an associative and unial ring such that for all $r, s \in \mathbb{K}$ and $a, b \in A$, $(ra)(sb) = (rs)(ab)$.

If every non-zero element $u \in A$ is invertible, then $A$ is a *division algebra ober* $\mathbb{K}$.

## 2.2 Linear Algebra over Division Algebras

## 2.3 Quaternions

## 2.4 Quaternionic Matrix Groups

## 2.5 Automorphism Groups of Algebras

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# Chapter 3

# Lie Algebras and Tangent Spaces

## 3.1 Lie Algebras

**Definition 3.1.** A *Lie algebra over* $\mathbb{K}$ consists of a vector space $\mathfrak{a}$ over a field $\mathbb{K}$, together with a bilinear map $[\,,] : \mathfrak{a} \times \mathfrak{a} \longrightarrow \mathfrak{a}$ called the *Lie bracket*, such that for $x, y, z \in \mathfrak{a}$,

$$[x, y] = -[y, x],$$
$$\text{and} \quad [x, [y, z]] + [y, [x, z]] + [z, [x, y]] = 0$$

Given two matrices $A, B \in M_n(\mathbb{K})$, thier *commutator* is the *Lie Bracket* is the matrix $[A, B] = AB - BA$. This defines a bilinear function

$$[\,,] : M_n(\mathbb{K}) \times M_n(\mathbb{K}) \longrightarrow M_n(\mathbb{K}); \quad (A, B) \mapsto [A, B]$$

which easily satisfies the definition. Hence $M_n(\mathbb{K})$ is a Lie algebra over $\mathbb{K}$ with the commutator as it's Lie bracket.

If all brackets in a Lie algebra are trivial, it is called *abelian*. We further devolop some of the basic algebraic notions of Lie algebra theory, analogous to the basic theory of groups and rings.

**Definition 3.2.** If $\mathfrak{a}$ is a Lie algebra over $\mathbb{K}$, then a subspace $\mathfrak{b} \subseteq \mathfrak{a}$ is a *Lie subalgebra* of $\mathfrak{a}$ if it is closed under the Lie bracket, that is, whenever $x, y \in \mathfrak{b}$, $[x, y] \in \mathfrak{b}$; we write $\mathfrak{b} \leqslant \mathfrak{a}$.

$\mathfrak{a}$ is a Lie subalgebra of itself, and so is $\{0\}$.

**Definition 3.3.** If $\mathfrak{a}$ is a Lie algebra over $\mathbb{K}$, then a vector subspace $\mathfrak{n} \subseteq \mathfrak{a}$ is a *Lie ideal* of $\mathfrak{a}$, given $[z, x] \in \mathfrak{n}$ for all $z \in \mathfrak{n}$ and $x \in \mathfrak{a}$; we then write $\mathfrak{n} \lhd \mathfrak{a}$.

A Lie ideal is *proper* if $\mathfrak{n} \neq \mathfrak{a}$, and is *non-trivial* if $\mathfrak{n} \neq 0$.

**Definition 3.4.** The *centre* of $\mathfrak{a}$ is the subset

$$z(\mathfrak{a}) = \{z \in \mathfrak{a} : \forall x \in \mathfrak{a}, [z, x] = 0\}$$

**Definition 3.5.** The *derived subalgebra* $\mathfrak{a}' \leqslant \mathfrak{a}$ is the vector subspace spanned by all the brackets $[x, y]$, $x, y \in \mathfrak{a}$.

**Proposition 3.1.** *$z(\mathfrak{a})$ and $\mathfrak{a}'$ are Lie ideals of $\mathfrak{a}$.*

**Definition 3.6.** A Lie algebra is simple if every proper ideal of the Lie algebra is trivial.

**Definition 3.7.** Let $\mathfrak{g}, \mathfrak{h}$ be Lie algebras. A linear transformation $\phi : \mathfrak{g} \longrightarrow \mathfrak{h}$ is a *homomorphism of Lie algebras* is for all $x, y \in \mathfrak{g}$, $\phi([x, y]) = [\phi(x), \phi(y)]$.

**Proposition 3.2.** *Let $\phi : \mathfrak{a} \longrightarrow \mathfrak{b}$ be a homomorphism of Lie algebras. Then ker $\phi \lhd \mathfrak{a}$ and there is an isomorphism im $\phi \cong \mathfrak{a}/$ker $\phi$.*

## 3.2 Curves and Tangent Spaces

Throughout this section, let $G \subseteq GL_n(\mathbb{K})$ be a matrix group.

**Definition 3.8.** The *tangent space* to $G$ at $U \in G$ is

$$T_U G = \gamma'(0) \in M_n(\mathbb{K}) : \gamma \text{ is a differentiable curve in } G \text{ with } \gamma(0) = U$$

**Proposition 3.3.** *$T_U G$ is a real vector subspace of $M_n(\mathbb{K})$.*

*Proof.* Suppose $\alpha, \beta$ are differentiable curves in $G$ for which$\alpha(0) = \beta(0) = U$. Then

$$\gamma : \text{dom } \alpha \cap \text{dom } \beta \longrightarrow G; \quad \gamma(t) = \alpha(t)U^{-1}\beta(t)$$

is also a differentiable curve in $G$ with $\gamma(0) = U$. Now,

$$\gamma'(t) = \alpha'(t)U^{-1}\beta(t) + \alpha(t)U^{-1}\beta'(t)$$

Hence $\gamma'(0) = \alpha'(0)+\beta'(0)$, which shows that $T_U G$ is closed under addition.

Similarly, if $r \in \mathbb{R}$ and $\alpha$ is a differentiable curve in $G$ with $\alpha(0) = U$, then $\eta(t) = \alpha(rt)$ defines another such curve. Since $\eta'(0) = r\alpha'(0)$, we see that $T_U G$ is closed under real scaler multiplication. ∎

**Definition 3.9.** The *dimension* of a real matrix group $G$ is dim $G = \dim_{\mathbb{R}} T_I G$. If $G$ is complex, then its *complex dimension* is $\dim_{\mathbb{C}} G = \dim_{\mathbb{C}} T_I G$.

We denote $T_I G = \mathfrak{g}$. In fact, $\mathfrak{g}$ has the algebraiic structure of a real Lie algebra.

**Theorem 3.1.** *If* $G$ *is a matrix subgroup, then* $\mathfrak{g} = T_I G$ *is a real Lie subalgebra of* $M_n(\mathbb{K})$*. If* $G$ *is complex matrix subgroup and* $\mathfrak{g}$ *is a subspace of* $M_n(\mathbb{C})$*, then* $\mathfrak{g}$ *is a complex Lie subalgebra.*

*Proof.* We will show that for differentiable curves $\alpha, \beta$ in $G$ which satisfy $\alpha(0) = \beta(0) = I_n$, there is another such curve $\gamma$ with $\gamma'(0) = [\alpha'(0), \beta'(0)]$.

Consider the function

$$F : \mathrm{dom}\ \alpha \times \mathrm{dom}\ \beta \longrightarrow G; \quad F(s,t) = \alpha(s)\beta(t)\alpha(s)^{-1}.$$

This curve is clearly continuous and differentiable with respect to each of the variables $s$ and $t$. For each $s \in \mathrm{dom}\ \alpha$, the function $F(s, \ ) : \mathrm{dom}\ \beta \longrightarrow G$ is a differentiable curve in $G$ with $F(s,0) = I_n$. On differentiating we find

$$\frac{d}{dt|_{t=0}} F(s,t) = \alpha(s)\beta'(0)\alpha(s)^{-1},$$

and so $\alpha(s)\beta'(0)\alpha(s)^{-1} \in \mathfrak{g}$. Since $\mathfrak{g}$ is a closed subspace of $M_n(\mathbb{K})$, whenever this limit exists, we also have

$$\lim_{s\to 0} \frac{1}{s}(\alpha(s)\beta'(0)\alpha(s)^{-1} - \beta'(0)) \in \mathfrak{g}.$$

Using the matrix rule for differentiating an inverse, we have

$$\begin{aligned}
\lim_{s\to 0} \frac{1}{s}(\alpha(s)\beta'(0)\alpha(s)^{-1} - \beta'(0)) &= \frac{d}{ds|_{s=0}} \alpha(s)\beta'(0)\alpha(s)^{-1} \\
&= \alpha'(0)\beta'(0)\alpha(0) - \alpha(0)\beta'(0)\alpha(0)^{-1}\alpha'(0)\alpha(0)^{-1} \\
&= \alpha'(0)\beta'(0) - \beta'(0)\alpha'(0) \\
&= [\alpha'(0), \beta'(0)]
\end{aligned}$$

Hence $[\alpha'(0), \beta'(0)]$ must be the form $\gamma'(0)$ for some differentiable curve. ∎

So for each matrix group $G$ there is a Lie algebra $\mathfrak{g} = T_I G$. A suitable type of homomorphism $G \longrightarrow H$ between matrix groups gives rise to a linear transfromation $\mathfrak{g} \longrightarrow \mathfrak{h}$ which is a homomorphism of Lie algebras.

**Definition 3.10.** Let $G, H$ be matrix groups, and $\varphi : G \longrightarrow H$ be a continuous map. Then $\varphi$ is said to be a *differentiable map* if, for every differentiable curve $\gamma : (a,b) \longrightarrow G$, the curve $\varphi \circ \gamma : (a,b) \longrightarrow H$ is differentiable and has derivative $(\varphi \circ \gamma)'(t) = \frac{d}{dt}\varphi(\gamma(t))$; and if two differentiable curves satisfy $\alpha(0) = \beta(0)$, and $\alpha'(0) = \beta'(0)$, then $(\varphi \circ \alpha)'(0) = (\varphi \circ \beta)'(0)$.

A differentiable map which is alsp a group homomorphism is called a *differential homomorphism*. A continuous homomorphism of matrix groups that is also differentiable is called a *Lie homomorphism*.

**Proposition 3.4.** *Let* $G, H, K$ *be matrix groups with* $\varphi : G \longrightarrow H$ *and* $\theta : H \longrightarrow K$ *differential homomorphisms. Then,*

(i) *for each* $A \in G$*, there is a real linear transformation* $d\varphi_A : T_A G \longrightarrow T_{\varphi(A)} H$ *given by* $d\varphi_A(\gamma'(0)) = (\varphi \circ \gamma)'(0)$*, for every differentiable curve* $\gamma : (a, b) \longrightarrow G$ *with* $\gamma(0) = A$*;*

(ii) *we have* $d\theta_{\varphi(A)} \circ d\varphi_A = d(\theta \circ \varphi)_A$*;*

(iii) *for the identity map* $\mathrm{Id}_G$*,* $d\mathrm{Id}_G = \mathrm{Id}_{T_A G}$*.*

**Theorem 3.2.** *Let* $G, H$ *be matrix groups and* $\varphi : G \longrightarrow H$ *be a differential homomorphism. The the derivative* $d\varphi : \mathfrak{g} \longrightarrow \mathfrak{h}$ *is a homomorphism of Lie algebras.*

*Proof.* Following the ideas and notations from the proof of theorem 3.1,for differentiable curves $\alpha, \beta$ in $G$ with $\alpha(0) = \beta(0) = I$, we can use the composite function $\varphi \circ F$ given by

$$\varphi \circ F(s,t) = \varphi(F(s,t)) = \varphi(]alpha(s))\varphi(\beta(s))\varphi(\alpha(s))^{-1}$$

t0 deduce that

$$d\varphi\left([\alpha'(0), \beta'(0)]\right) = [d\varphi(\alpha'(0)), d\varphi(\beta'(0))].$$

■

## 3.3 Lie Algebras of Some Matrix Groups

Here we determine the Lie algebras of some important familes of real and complex matrix groups.

1. **General and Special Linear Groups**
2. **Upper Triangular and Unipotent Groups**
3. **Affine Groups**
4. **Orthagonal and Special Orthangonal Groups**
5. **Unitary and Special Unitary Groups**
6. **Quaternionic Symplectic Group**

## 3.4 $SO(3)$ and $SU(2)$

The lie algebras $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$ are both 3-dimensional vector spaces, for example with the following bases:

$$\mathfrak{so}(3): \quad P = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$\mathfrak{su}(2): \quad H = \frac{1}{2}\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad E = \frac{1}{2}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad F = \frac{1}{2}\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

The non trivial Lie brackets among these are,

$$[P, Q] = R, \; [Q, R] = P, \; [R, P] = Q,$$

$$[H, E] = F, \; [E, F] = H, \; [F.H] = E.$$

This implies that the real linear isomorphism $\varphi : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$;

$$\varphi(xH + yE + zF) = xP + yQ + zR \quad (x, y, z) \in \mathbb{R}$$

satisfies $\varphi([U, V]) = [\varphi(U), \varphi(V)]$, so is an isomorphism of real Lie Algebras. Thus these Lie algebras look the same algebraically. This suggests that there might be a close relationships between the groups themselves. Notice also that both $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic to $\mathbb{R}^3$ with the vector cross product as its Lie bracket.

▲▼▲

# Chapter 4

# Lie Groups

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# Chapter 5

# Homogenous Spaces

# References

[Baker, 2002] Baker, A. (2002). *Matrix Groups*. Springer Undergraduate Mathematics Series. Springer-Verlag.

[Trapp, 2005] Trapp, K. (2005). *Matrix Groups for Undergraduates*. Student Mathematical Library. American Mathematical Society, second edition.